



**Middlesex
University
London**

Data-Driven Inventory Management: Amazon Case Study

**by Amir Lorvand
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1. Introduction

Amazon, the global leader in e-commerce and cloud computing is a prime example of effective data use in decision-making and operational optimisation (Jeble et al, 2017). Among the many fields, Amazon leverages data in, one sector that really stands out to be dependent on it is inventory management (Aghababaei, 2024). Proper inventory management enables product availability, minimises cost, and enhances customer experience (Jeble et al, 2017). This paper focuses on a case study of Amazon to analyse how it employs internal and external data sources to make optimal inventory decisions.

Amazon performs millions of transactions daily based on its reach to the international markets resulting in huge volumes of data. When broken down, it gives valuable information on customer preferences, the trends happening in the market, and the best way to manage the supply chain (Chen et al. 2019). The company has a vast network of fulfillment centers around the world and suppliers; there is always the middle of how to ensure that the right products get to the right place at the right time (Chen et al. 2019). To address these challenges, Amazon has built a strategic BI model that incorporates data warehousing, Big data processing tools, and cloud computing services technologies to address large datasets (Chen et al. 2019).

In order to ensure that the inventory levels are as precise as possible, during periods when there is usually a high traffic of customers for example, Black Friday or Prime Day, Amazon pulls data from the platform in real-time to determine precise inventory dynamics (Hewage et al., 2018). This allows the company to know when the demand rate increases, and avoid situations where there are no products to supply customers due to an increase in shopping rates (Hewage et al., 2018). Structured data includes company sales transactions

while unstructured data comprises data like customer feedback and suppliers' information (Zhang et al., 2020).

Specifically, this report evaluates the current Amazon stock inventory decision-making structures by focusing on data storage forms such as data warehouse, Big Data, and Cloud. It then gives the most advantageous data strategy for Amazon, in addition to easy operation and integration.

2. The Data Requirements

Amazon has a warehousing and logistical problem inherent with having a large amount of stock; to deal with this, the company has accumulated and processed information from various internal and external sources. These data types can enhance stock data and boost decision-making activities.

Internal Data:

Internal data comes from the day-to-day operations in an organization, such as:

- **Sales Transactions:** Volume, frequency, and regional data that capture the rate of sale in actual time are an aspects that Amazon does track for products. For instance, during Prime Day, actual selling data is utilised to replenish inventory to the degree necessary. This data is stored in AWS services like Amazon Redshift and Amazon S3.
- **Customer Behavior:** Amazon does not only capture and review history and past browsers and searches as well as item and basket data with a view to anticipating demand for products and providing customers with the right recommendations thus improving on satisfaction levels. This data is stored in a comparatively simpler form in data lakes by utilising Amazon S3 and is analysed

using AWS Glue or Amazon EMR.

- **Warehouse and Logistics Data:** Inventory levels, warehouse storage capacity, and the estimated time of delivery are well followed in order to fine tune the logistics of the overall supply chain. The details of the warehouse are kept in the Amazon DynamoDB.

External Data:

External data is collected and gathered from outside of the organization, like:

- **Competitor Pricing:** Through competitor monitoring, Amazon ensures they maintain a competitive edge with the use of real-time pricing incorporated in the DPS to provide the best offer to customers. It is stored in Amazon Redshift.
- **Supplier Data:** In this case, the supplier stock, production time, and delivery time are tracked so as to avoid inconveniences such as lack of stocks or delay in production and delivery. It is kept through the use of Amazon RDS and Redshift for relational database information.
- **Market Trends and Economic Data:** Market research and economic analytics help identify demands for the goods, and the stocks are changed when consumer behavior shifts, for example, during the pandemic. Big data outside ArcGIS is stored on Amazon S3 and processed by AWS Lambda.

These internal and some external data sources give Amazon regular and broad views on the company's internal situation and the external environment for effective and efficient decision-making in the inventory management.

3. The Data Storage

In more detailed classification structures, Amazon uses data superseries, data warehouses, BigData systems, and other cloud storage approaches that are proper in structured and unstructured data to make the right and informed decision-making about the inventory, customer behaviors, and the supply chain (Pandis, 2021).

Data Warehousing

Structured data like sales, orders of customers, and inventory levels are stored in data warehouses that are important to Amazon. Amazon Redshift is an important tool that operates in the cloud environment for large amounts of data (Thuraga, 2024). Thanks to Redshift, Amazon can manage and process petabytes of the structured data while making fast queries, which are especially critical in the moments when the firm needs to make decisions rapidly, for example, during Black Friday (Thuraga, 2024).

Advantages of Data Warehousing

- **Efficient Querying:** Redshift helps Amazon to produce big data on the sales volume, inventory, and customers, which can help Amazon make necessary changes during surge demand.
- **Historical Data:** Serves as a warehouse of prior data, used for identification of past trends, which are indispensable for future demand prediction.
- **Data Consistency:** Makes sure that structural data is credible and conducive to accurate reporting.

Nevertheless, data warehouses cannot be used for loosely structured data such as customer feedback or posts on social networks, and this is where Amazon uses Big data systems.

Suitable Data Schema: Star Schema

Currently, Amazon Redshift that is a data warehouse based on the structure of Redshift uses the star schema (Parchas et al., 2020). This is a star-like schema where a large central fact table of quantitative data such as sales figures is surrounded by dimension tables such as products, time, and customers (Parchas et al., 2020).

The fact table as can be seen in Figure 1, consists of transactional data such as total amount of sales or total revenue whereas the dimensions enable Amazon to carry out sale analysis under different parameters (Parchas et al., 2020).

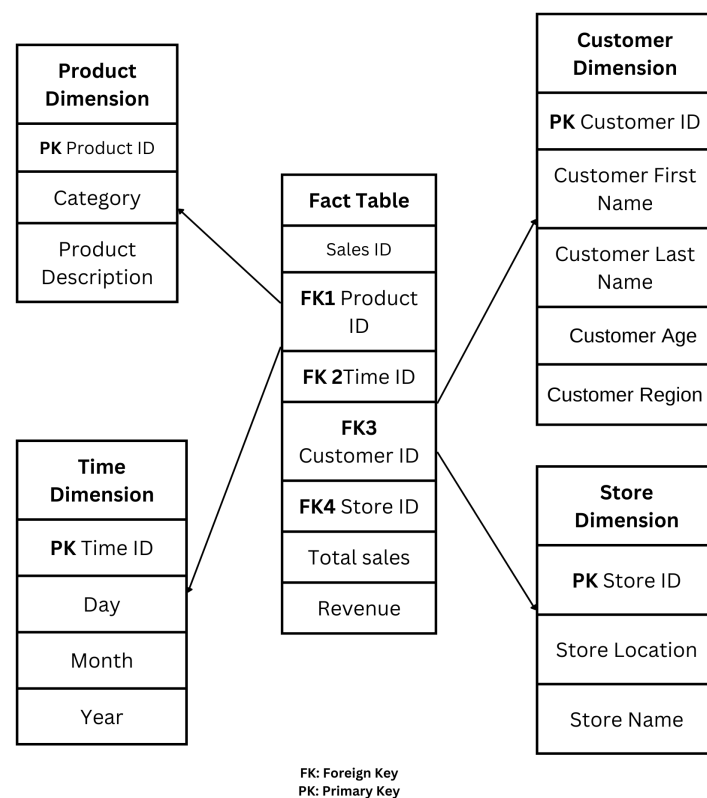


Figure 1: Star Schema Model

- A product dimension table could contain product identification numbers, categories and descriptions.
- Data is divided between the tables by time intervals: daily, monthly and yearly.
- The customer dimension table collects the basic customer demographics such as age, region, etc.
- The store dimension table is used to record information regarding the geographical location and name of the stores through which sales were made in order for Amazon to assess the result on a store level.

Advantages

- **Simplified Queries:** Allows for ease creating reports such as; sales region wise report or inventory by product type reports.
- **Optimised for Fast Retrieval:** Most suited for Amazon's reporting since it offers a fast summaries' generation during periods of high demand.

Disadvantage

- **Data Redundancy:** This means that in the star schema, the dimension tables may repeat the same data which may add to storage capacity in large dataset and complexities.

ETL Process (Extract, Transform, Load)

Amazon's ETL is actually very crucial for transforming raw data from different sources into data warehouses (Redshift) and big data platforms (Hadoop, Spark) of the organisation Nguyen, 2023). It means that the data collected has been pre-processed, reformed and made suitable for usage in making decisions.

1. Extraction

Amazon extracts data from:

Internal Sources:

- **E-commerce platform:** Sales transactions, customer orders, and inventory data.
- **Supplier systems:** Stock levels, supplier orders, and shipping updates.

External Sources:

- **APIs:** Market trends, competitor pricing.
- **Social Media:** Customer reviews and sentiment from platforms like Twitter.

Techniques:

- **APIs** for real-time data, and web scraping for collecting unstructured data like competitor pricing.

2. Transformation

The transformation stage ensures consistency by:

- **Data Cleaning:** Removing duplicates and correcting errors (e.g., invalid addresses).
- **Standardisation:** Ensuring data from multiple sources follows the same format.
- **Integration:** Merging data from different sources (e.g., combining sales with inventory data for performance reports).

Techniques:

- **Data mapping** aligns data from various sources, ensuring consistency.

3. Loading

- **Structured Data:** Loaded into Redshift for querying (e.g., sales reports by product or region).
- **Unstructured Data:** Loaded into Hadoop/Spark for real-time analysis (e.g., analysing customer sentiment).

Techniques:

- **Batch loading** for daily sales data into Redshift.
- **Real-time streaming** for unstructured data (e.g., clickstream data) into Hadoop.

The efficient and robust ETL procedure for Amazon helps make its data trustworthy and obtainable for analysis to assist with stock, selling, and consumers' assessments (Nguyen, 2023).

Online Analytical Processing (OLAP)

Amazon employs OLAP operations when drawing conclusions as well as when categorising products, locations and temporal periods for quick management of sales and inventory, through tools such as slicing, dicing, rolling up, rolling down and pivoting (Kulkarni, 2023).

Slicing and Dicing

Slicing lets Amazon peer into the different data parts relevant to specific events like UK TV sales during Black Friday to evaluate its products' performance in the area (Kulkarni, 2023).

Dicing allows Amazon to look at data from different angles, for instance, with a glance at the sales of both TVs and smartphones in North America and Europe

during Black Friday. However, it gives a wider comparison on the performance within the regions and on the various products (Figure 2) (Kulkarni, 2023).

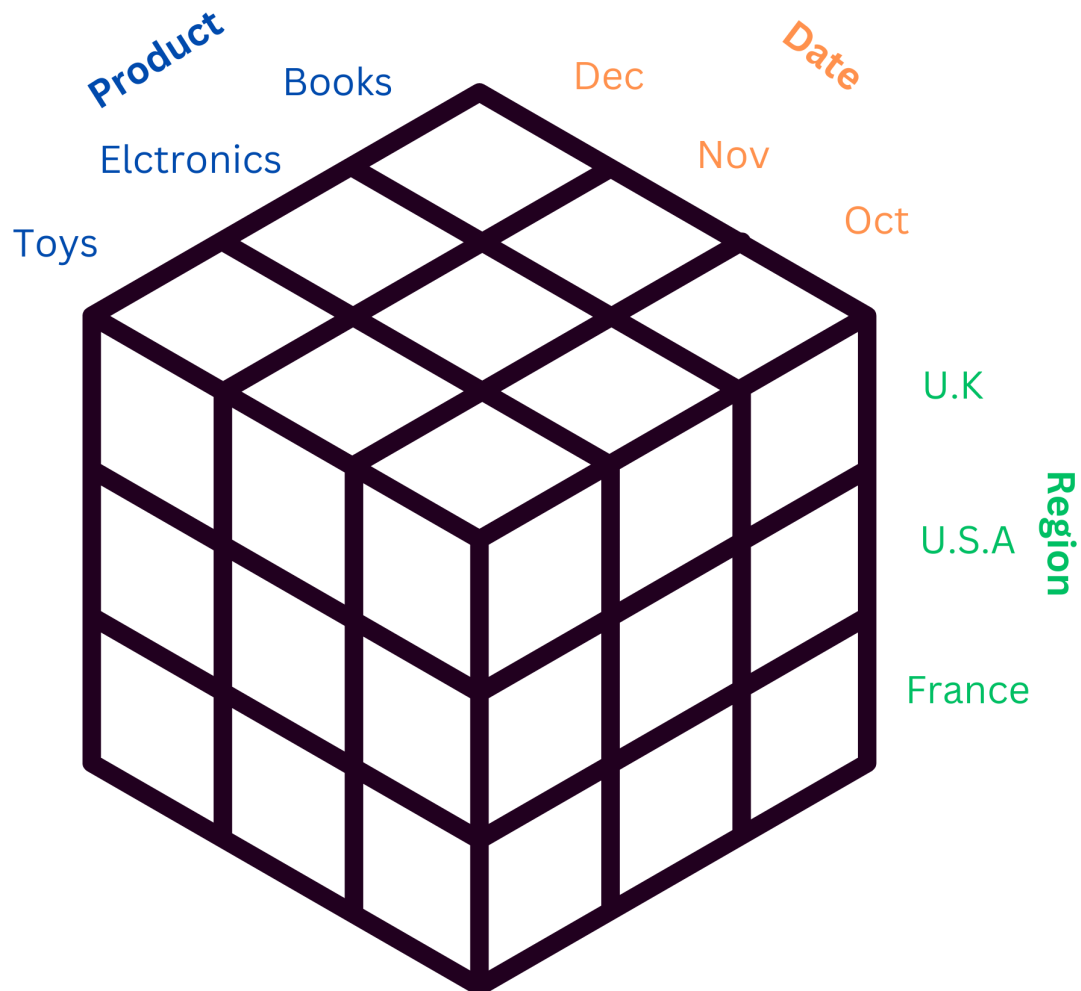


Figure 2: Data Cube

Roll-Up/Roll-Down

Roll-up is used to bring data together for additional level information (Kulkarni, 2023). For example, sales data can be grouped at Amazon in order to observe

monthly patterns in the UK, which would provide a general idea of the most popular products.

Roll-down extends to more specifics, for instance, when looking at the sales made over Prime Day, one may want to know where the particular city or fulfillment center was raking in the most sales (Kulkarni, 2023).

Pivoting

Pivoting enables Amazon to rotate views in-analyse, for instance, sales across regions or sales across products. This aids in pattern recognition regarding some products' performance in some regions.

Example Query Using OLAP

According to Figure 2, for the query, "Which product categories performed best in the UK during Black Friday?", Amazon could:

- Filtration of the UK sales data based on Black Friday.
- Dice to evaluate prospects 'performance in diverse product categories.
- Total of sales across all the continents to be rolled up.
- Roll down the above table to drill down to specific cities for further analysis.
- Switch between sales by product category and geography.

These OLAP operations enable Amazon to analyse and make decisions on stocks, sales and customers' trends.

Big Data Systems

There are numerous types of information, including customer reviews, posts of a social nature, and clickstreams, that are difficult to handle in a data warehouse, but Amazon has to process them.

Big Data and the 3Vs

- **Volume:** Amazon creates huge volumes of data every day in terms of reviews and clickstreams.
- **Variety:** The data in the work comes in text format like reviews, images like product photos, and clickstream data.
- **Velocity:** It remains important in real-time processing for clickstreams, insights gleaned are often useful and need to be acted upon instantly for recommendations for a product or a marketing campaign.

Why Big Data Cannot Be Stored in a Data Warehouse

1. **Volume and Velocity:** As for other data storage systems such as Redshift, they hardly meet the large-scale and real-time requirements of unstructured data.
2. **Variety:** Data in the form of text and multimedia from the various reviews and social media posts do not conform well to the rigidity of a data warehouse model of the actual business.
3. **Real-Time Analysis:** Systems such as Hadoop architecture perform the real-time processing required for dynamic data as found with clickstreams.

Hadoop Framework for Big Data

Amazon uses **Hadoop** to collect, store, and analyse its unstructured data:

- **HDFS (Hadoop Distributed File System):** HDFS divides the big data into chunks, for instance customer reviews and then store these chunks in different servers (Borthakur, 2007). Amazon uses this distributed storage and compute system to provide a massive distributed storage to accommodate the use of the immense customer feedbacks and interactions (Vishal Reddy et al., 2022).

- **MapReduce:** This processing model splits big analytic work into several tasks. For instance, MapReduce can transpose customer sentiment on the number of product reviews, which may reach millions in a day (Hashem et al., 2023). Part of the reviews is analysed by a single server, and then the results are merged into the final sentiment analysis.
- **Apache Spark:** Hadoop is combined with Spark for the processing of real-time data, such as clickstream activity.

Advantages

- **Scalability:** Easily handles increasing data.
- **Real-Time Analytics:** Provides instant insights.
- **Flexibility:** Handles diverse data formats.

Disadvantages

- **Resource-Intensive:** Requires significant computational resources.
- **Complex Management:** Needs technical expertise and specialised tools.

Cloud Storage

Essentially, Amazon uses AWS whose main function is to store its cloud data in both structured and unstructured formats (Witting, 2023). Cloud computing helps Amazon in managing its data through Web technologies rather than the physical infrastructure and provides facilities for rental of computing resources (Witting, 2023).

Cloud Computing in General

Cloud computing provides on-demand computing solutions such as storage, processing and data management over the Internet in a prosperous way thereby helpful for organisations like Amazon who analyse huge amount of data all around the world (Sunyaev, 2020). The following image (Figure 3) shows the cloud storage architecture:

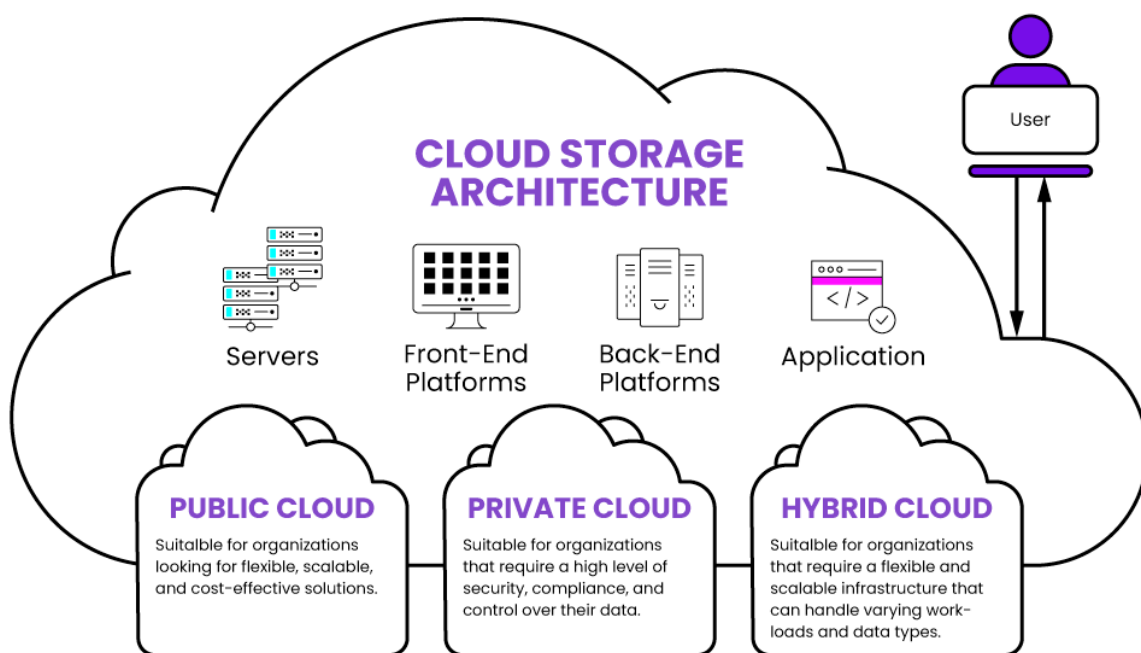


Figure 3: Cloud Storage Architecture (Weka.io, 2024)

Advantages of Cloud Storage

- **Scalability:** Easily adjusts to demand, like during Black Friday.
- **Cost Efficiency:** Pay only for what's used.
- **Global Access:** Real-time data access worldwide.
- **Security:** Strong encryption and regulatory compliance.

Disadvantages of Cloud Storage

- **Security Risks:** Potential for cyberattacks.
- **Internet Dependency:** Disruptions can affect access.

Cloud Storage for Data Warehousing

Amazon's data is kept in Amazon Redshift which is a cloud data warehouse and adapted for structured data like sales and inventory (Gupta et al., 2015). Less dependence on physical or remote computing is required for servers which in turn is more flexible and cheaper to implement for structured data management for Amazon (Gupta et al, 2015).

Cloud Storage for Big Data Systems

With the large data format in the form of customer feedback and clickstream data, Amazon has adopted S3, and it also assimilates it with other big data systems, including Hadoop and Spark (Priyadarshini and Veeramanju, 2022). The fundamental of this cloud structure empowers Amazon to archive and compute vast structures of obstinate data alongside sufficiently capturing the elasticity required in analysing real time data (Priyadarshini and Veeramanju, 2022).

4. Conclusion and Recommendations

In this report, we sought to understand how Amazon exploits internal and external data sources to make inventory decisions. Much information that Amazon gathers is collected during the course of its business manipulations, such as micro/macro sales data, customer activity, data on warehousing and logistics, competitor prices, information about suppliers and markets. These data types differ in terms of organisation where sales transactions, and identification of suppliers, as well as inventory details can be structured data stored in some data warehouse, such as Amazon Redshift. On the other hand, from the data type perspective there are structured data sources like customer databases, transaction records, etc., unstructured data like customer reviews, social media interactions, click stream data, etc., and semi structured data sources like text messaging, emails, etc., Which is managed through big data tools like Hadoop and Spark.

As Amazon comes across both structured and unstructured data, a hybrid system must be applied. Traditional data must be stored and analysed in a data warehouse and new info-tender in big data environments. It has made Amazon a flexible combination and hold all types of data handy for advanced form of analytics, real time insights and accurate forecasts.

Moreover, working with AWS and deploying cloud storage is useful for using in this hybrid model as well. The use of cloud storage means that Amazon is able to accommodate large volumes of data and can easily scale this up in terms of storage space depending on the current needs at any given time, and specifically during some of the biggest occasions such as the Black Friday sales.

Therefore, the integration of DW, BDA and cloud storage will help Amazon to extract the highest worthy of its data to fuel its decision-making process and enhance its operation.

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